

Spin Dynamic Methods

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1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, time is measured in seconds, and magnetic fields $\mathbf{H}$ are expressed in A m^{-1} . The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

$\boldsymbol{\sigma}$ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both have field units.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned}\gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}.\end{aligned}$$

Convention warning. Some references define the effective field as $\mathbf{B}_{\text{eff}}$ in tesla and use $|\gamma|$, whereas this library uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\gamma_0 = \mu_0 |\gamma|$. Check this convention before copying an equation into simulation code.

2 Spin Hamiltonians

2.1 Atomistic Spin Hamiltonian

Equation. Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned}\mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\ & - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}.\end{aligned}\tag{2.1}$$

Definitions. J_{ij} is the exchange-link energy in the ASTRO sign convention: $J_{ij} < 0$ favors parallel alignment, whereas $J_{ij} > 0$ favors antiparallel alignment. K_i is the uniaxial anisotropy energy, $\hat{\mathbf{e}}_i$ is the easy-axis direction, $\mathbf{D}_{ij}$ is the Dzyaloshinskii–Moriya vector, and $\boldsymbol{\mu}_i$ is the magnetic moment at site i .

Assumptions. Each $\mathbf{s}_i$ is a classical unit vector. The pair coefficients satisfy $J_{ji} = J_{ij}$ and $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. The factors of $1/2$ assume that the pair sums include both ordered pairs (i, j) and (j, i) . If each physical bond is instead included only once, for example in a unique-bond sum $\sum_{\langle i, j \rangle}$, both factors of $1/2$ must be omitted. These are equivalent pair-counting conventions and must not be mixed.

Pair counting and effective field. Let $\mu_i = |\boldsymbol{\mu}_i|$ and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}.\tag{2.2}$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}.\tag{2.3}$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of $1/2$. Consequently, no extra factor of $1/2$ belongs in either effective-field expression.

Units. All terms in $\mathcal{H}$ have units of energy. Here the external flux density $\mathbf{B}_{\text{ext}}$ is measured in tesla, so no additional factor of μ_0 appears in the Zeeman term.

Source. For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

Implementation notes. The ASTRO implementation evaluates the field, $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$, rather than the total energy and includes both forward and backward nearest neighbors. The J_{ij} used above is therefore the coefficient stored in ASTRO's exchange-link arrays: its exchange field is $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$ [4]. No factor of $1/2$ should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO's DMI component expressions correspond to $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$ and the antisymmetry $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ used above.

3 Magnetization Dynamics

LLGS form with field-like torque.

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (3.1)$$

Equivalent LL form with field-like torque.

$$\begin{aligned} (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha \gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) \\ & + \alpha \gamma \tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \alpha \gamma \tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (3.2)$$

Torque amplitudes. For a ferromagnetic layer,

$$\tau_{\text{DLT}} = \frac{\hbar \theta_{\text{SH}} J_c}{2e t_{\text{FM}} M_s}, \quad \tau_{\text{FLT}} = \beta \tau_{\text{DLT}}.$$

Torque-sign convention. In the convention of Eq. (3.1), the sign product of the damping-like and field-like SOT is negative when $\beta > 0$ [1]. Specifically, the scalar prefactors of their respective vector structures are $-\gamma \tau_{\text{DLT}}$ and $+\gamma \tau_{\text{FLT}}$, so their sign product is

$$(-\gamma \tau_{\text{DLT}})(+\gamma \tau_{\text{FLT}}) = -\gamma^2 \beta \tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive β means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes τ_{DLT} and τ_{FLT} themselves have the same sign.

4 Thermal Stability and Fluctuations

4.1 Thermal Stability Factor

Equation. For a single energy barrier E_b , the thermal stability factor is

$$\Delta = \frac{K_u V}{k_B T} \quad (4.1)$$

Units. E_b and $k_B T$ must use the same energy unit; Δ is dimensionless.

4.2 Thermal Field in a Macrospin Model

Correlation. For the $\mathbf{H}$ -field convention used in Eq. (3.1), one common white-noise convention is

$$\langle H_{\text{th},i}(t) H_{\text{th},j}(t') \rangle = \frac{2\alpha k_B T}{\gamma_0 M_s V} \delta_{ij} \delta(t - t'). \quad (4.2)$$

Implementation notes. The discrete-time variance depends on the stochastic calculus convention, the form of the LLG equation, and the time-step definition. Derive the sampled-noise amplitude from the exact solver convention rather than copying Eq. (4.2) without checking it.

References

- [1] J. Park, G. E. Rowlands, O. J. Lee, D. C. Ralph, R. A. Buhrman, Macrospin modeling of sub-ns pulse switching of perpendicularly magnetized free layer via spin-orbit torques for cryogenic memory applications, *Appl. Phys. Lett.*, **105**, 102404 (2014).
- [2] R. F. L. Evans, W. J. Fan, P. Chureemart, T. A. Ostler, M. O. A. Ellis, R. W. Chantrell, Atomistic spin model simulations of magnetic nanomaterials, *J. Phys.: Condens. Matter*, **26**, 103202 (2014).
- [3] Z. Kong, Z. Xu, X. Zhang, B. Zhao, Y. Yang, L. Zhu, Z. Zhu, Spatially asymmetric switching under spin-orbit torque and Dzyaloshinskii–Moriya interaction, *J. Appl. Phys.*, **140**, 063903 (2026).
- [4] Z. Zhu, ASTRO atomistic spin-dynamics code, GitHub repository, commit `9c807cf`.